



University of Essex

Title:

**Portfolio Optimisation using Linear Regression and CAPM:
An Evaluation of Methodology and Results**

Subject:

Report

Submitted as part of:

CF969-7-SP/ZU: Machine Learning for Finance Assignment 1

Module Supervisor:

Panagiotis Kanellopoulos

Author:

Victoria Grace Oghuan (Registration: 2400724)

1.Introduction

This report details the evaluation and interpretation of portfolio optimisation using the Capital Asset Pricing Model (CAMP) to estimate the risk and expected returns of 10 assets over a five-year period. Portfolio Optimisation is a key concept in finance, it's a process of creating a portfolio of investments by selecting the best combination of assets with the goal of achieving the maximum returns with minimum risk of investment for a target return. It involves applying quantitative methods and analysis of variables related to the volatility, correlation, and to predicted returns and risks for portfolio optimisation. Investment decisions are made across the assets in a calculated way by applying statistical and mathematical models CAMP, Linear Regression, Covariance Matrix, Quadratic Programming and Efficient Frontier for visualisation. This report details the methodology, evaluates the results, and interprets key findings.

2.Methodology

2.1 Data Collection and Daily Returns

The historical stock price data for 10 assets (AAPL, MSFT, GOOGL, AMZN, TSLA, META, NFLX, NVDA, JPM, V) were downloaded from Yahoo Finance in addition to the S&P 500 index (^GSPC) over a 5-year period from January 2019 to December 2023. The daily returns for each asset were calculated based on the closing price, where daily returns are the percentage change in the asset's price from one trading day to the next.

```
assets_daily_returns = data.copy().pct_change().dropna()
```

In the code used to calculate the daily asset returns, the Pandas method **pct_change()** computes the daily percentage change from the previous row, and **dropna()** handles any NaN values, particularly the first row, as there is no previous row exists before the start date.

2.2 CAPM based Linear Regression for expected returns and risk

The Capital Asset Pricing Model (CAPM) is a financial model that calculates the estimated expected returns and risk for each asset via linear regression. The CAPM formula is given by:

$$R_i - R_f = \alpha_i + \beta_i(R_m - R_f) + \epsilon_i$$

Where R_i is the asset returns, R_f is the risk free rate (at 2% per annum), and it's calculated to the daily rate by dividing by 252 trading days, R_m the market return represented by (S&P 500), α_i is the excess returns and β_i measures the systematic risk, and ϵ_i is the error term. The α (alpha) is the excess returns in relation to the predicted beta, the beta coefficient is an asset's sensitivity to the overall market trends.

```
for asset in assets:
    alpha, beta, std_err, r_value, p_value = linregress(market_returns.values.squeeze(), assets_returns[asset].values)
```

The code loops through each asset in the assets list. Linear regression was performed for each asset using the **linregress** function from the **SciPy. Stats** library to calculate the alpha and beta coefficients. The **squeeze** call ensures the data is in the 1D array format by removing additional dimensions.

The expected returns were calculated from the estimated beta values using the following CAPM formula:

$$\mu_i = R_f + \beta_i(E[R_m] - R_f)$$

Where the average market returns over the 5-year period was ($E[R_m]$)

The variance of the residuals was calculated to measure the idiosyncratic risk of each asset, which is dependent on the individual asset's risk and independent of the market movements. This method separates risk into systematic risk (beta) and idiosyncratic risk, giving a risk profile for each asset.

2.3 Portfolio Optimisation

The Covariance Matrix was constructed to evaluate the portfolio risk considering the systematic and idiosyncratic risk of each asset. It depicts the relationship between the assets in the portfolio and consists of diagonal and off diagonal elements.

The systematic risk is calculated using the estimated beta values and market variance:

$$\sigma_{ij} = \beta_i \beta_j \sigma_m^2 \text{ (for } i \neq j \text{)}$$

The idiosyncratic risk is added to the diagonal elements:

$$\sigma_{ii} = \beta_i^2 \sigma_m^2 + \text{Var}(\epsilon_i)$$

The code iterates through all assets in the double loop **for i in range(n_assets): for j range(n_assets):** to compute the diagonal elements, such as the systematic and idiosyncratic risk and the covariance/relationship between the assets using the beta values and market variance as seen in the code in the .py file. The values are converted to **float** to prevent dimension errors and data type issues, ensuring capabilities with the libraries. Where **Sigma** is a square matrix holds the covariance values between assets

Following the Covariance Matrix the portfolio optimisation is expressed as Quadratic Programming (QP) problem addressing where the aim is to minimise risk subject to the following constraints. The optimisation problem is given by:

Objective Function and Code:

$$\text{Minimize } \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w}$$

```
portfolio_risk = 0.5 * (x @ Sigma @ x)
m_ef.setObjective(portfolio_risk, GRB.MINIMIZE)
```

Subject to the following constraints and generated code:

1. **Full Capital Allocation:** $\mathbf{w}^T \mathbf{1} = 1$, `m_ef.addConstr(x.sum() == 1, 'budget')`
2. **No Short Selling:** $w \geq 0$, `x = m_ef.addMVar(n, lb=0, name='weights')`

3. Target Return $w^T \mu = \mu_p$,

```
m.addConstr(gp.quicksum(mu[i] * x[i] for i in range(n)) == target_return, 'target_return')
```

The target return is chosen from a range in the middle of the minimum and maximum expected returns the code is generated as follows:

```
min_return = min(mu)
max_return = max(mu)
target_return = (min_return + max_return) / 2
```

The Gourbi model is created, and the portfolio weights are defined taking into account the constraints while minimizing risk and meeting the target return.

2.4 Efficient Frontier

The Efficient Frontier was generated by solving this quadratic programming problem for a range of 10 target return values spaced equally between the maximum and minimum expected returns.

Using `np.linspace(min_return, max_return, 10)` ensures that the 10 target returns are actually in a reasonable range dependent on expected returns gives 10 evenly spaced values between `min_return` and `max_return`. The efficient frontier shows the minimum risk (standard deviation) for each target return.

3.Results and Interpretation

3.1 Linear Regression Scatter Plots

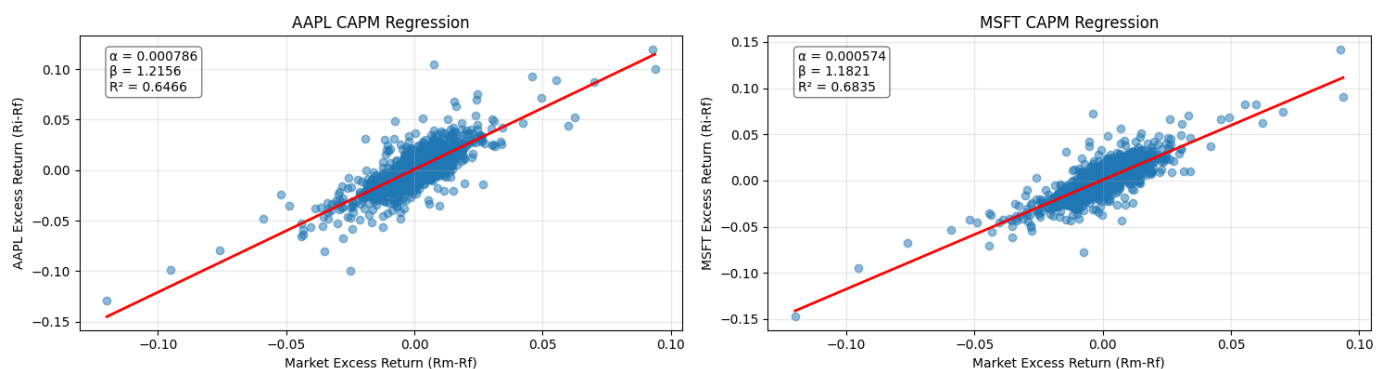


Figure 1:

The scatter plot for the 10 assets was generated however for display purposes two assets are shown above. The data points represent distribution of the daily excess returns where alpha is the intercept and beta are the slope of the regression line. The x axis is the market excess return the returns above the risk free rate given at 2 %. On the y-axis the individual assets excess return above the risk free rate. The r^2 value is between 0 and 1 indicating the assets dependent or independent of the market trends, The data points close to line of best fit suggests a strong relationship between the assets excess market return and market excess return indicating a good CAPM model fit.

3.1 Linear Regression Table

	Asset	Alpha	Beta	Expected Return	Idiosyncratic Risk
0	AAPL	1.215625	0.000786	0.000080	0.000412
1	MSFT	1.182136	0.000574	0.000080	0.000369
2	GOOGL	1.135638	0.000304	0.000080	0.000401
3	AMZN	1.065801	0.000151	0.000079	0.000492
4	TSLA	1.503262	0.001946	0.000080	0.001660
5	META	1.288666	0.000395	0.000080	0.000755
6	NFLX	1.042221	0.000291	0.000080	0.000846
7	NVDA	1.728839	0.001684	0.000080	0.001063
8	JPM	1.098643	0.000097	0.000079	0.000403
9	V	1.048806	0.000091	0.000079	0.000312

Figure 2: The table displays the key regression results for the 10 assets including alpha, beta, expected returns and idiosyncratic risk

Alpha represents the assets excess returns a measure of the asset's performance in comparison to the market, independent of market trends. Beta is the sensitivity of an asset to the overall market trends where $b=1$ moves in line with the market $b>1$ high volatility more sensitive to the market trends $b<1$ lower volatility and less sensitivity to the market.

All of the 10 assets have a positive alpha value, indicating historically they have all on average outperformed the market (S%P 500). NVDA has the highest alpha at 1.728839 indicating the highest excess returns. Amazon (AMZN) has the lowest alpha at (1.065801) indicating the lowest excess returns in comparison to the assets. Tesla (TSLA) at 0.001946 and NVIDIA Corporation (NVDA) at 0.001684 have the highest beta values indicating they are the most volatile assets and sensitive to the overall market trends. Visa Inc (V) at 0.000091 and JPMorgan (JPM) at 0.000097 have the lowest beta values indicating lower volatility and less sensitivity to market trends.

To summarise, NVDA and TSLA have the highest alpha and beta values they are the assets that have the greatest volatility, idiosyncratic risk and excess returns. V and JPM have the lowest beta and idiosyncratic risk values indicating they are more stable and less sensitive to market trends and less company specific risk.

3.2 Optimal Portfolio Weights

AAPL	0.239705
MSFT	0.187784
GOOGL	0.078407
AMZN	0.020676
TSLA	0.157013
META	0.058522
NFLX	0.035083
NVDA	0.210850
JPM	0.006443
V	0.005518

Figure 3: The table represents the optimal portfolio weights for each the 10 assets.

This table shows the optimal portfolio weight for each asset which is the percentage of the total investment that should be allocated across the 10 assets. This portfolio has an expected return of 0.000080 and a risk of 0.011453. AAPL and NVDA have the highest portfolio weights at approx. 24% and 21% & respectively. Together they make up almost half the portfolio approx. 43% this suggests these assets an investment taking on a higher risk for higher returns. However significant market fluctuations could cause as significance loss in returns the combination of the other assets in the portfolio allows for some level of diversification to manage risk.

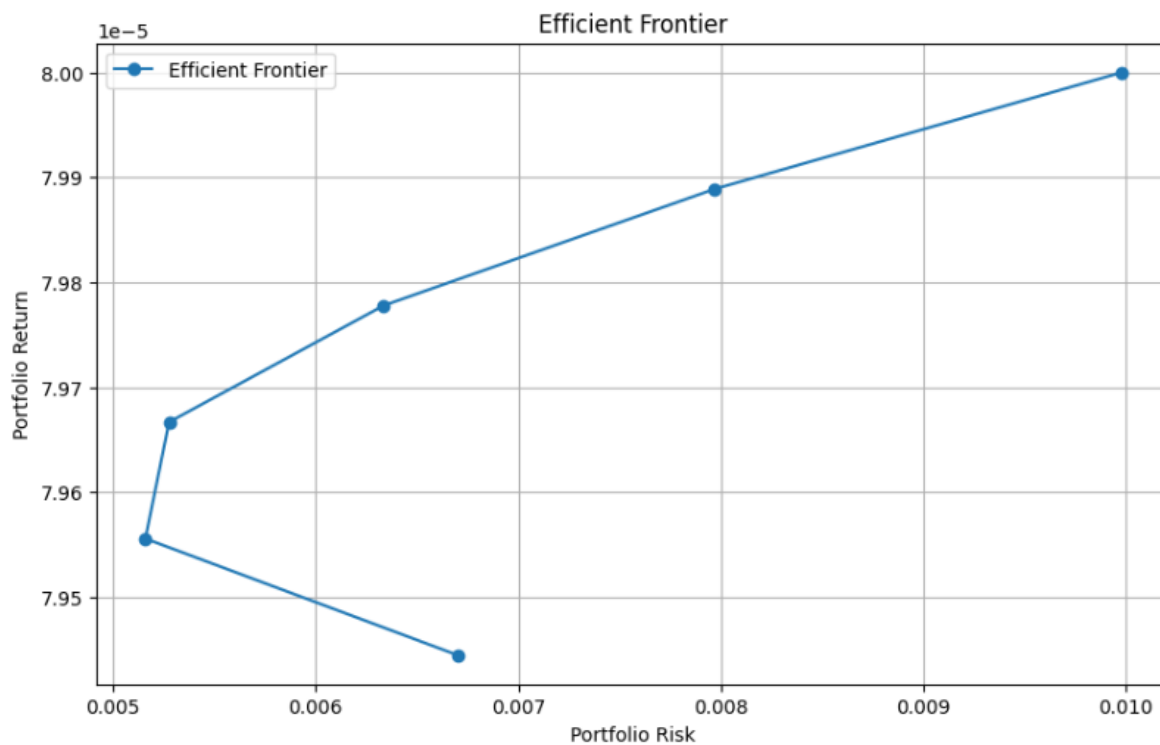


Figure 4: The efficient frontier a concave curve representing the relationship between portfolio return and portfolio risk.

The efficient frontier displays the relationship and trade-off between the portfolio return and portfolio risk measured by standard deviation. The graph shows that the relationship is not linear as seen by the concave shape however as expected as returns increase risk also increases.

Notably, around the 0.006 portfolio risk level and 7.96 portfolio return there's a significant change in the risk to return trade off this point is called "knee". To the left of the knee an increase of the portfolio returns from 0.005 to 0.006 yields an increase in risk from 0.004 to 0.005. By taking an a relatively moderate increase of risk of 0.001 an increase in returns of 0.00002 can be made. However, on the right side of the knee making the same increase in returns of 0.00002 requires a greater risk of 0.002,

Diversification is essential to the efficient frontier the combination of the assets with different portfolio weights and risk and returns creates a portfolio which minimises risk for the target return to make the best investments on assets.

4. Conclusion

In conclusion the report has shown the application of portfolio optimisation methods to real world financial applications ultimately building the optimal portfolio given the specified constraints and target returns.